

EX:	
DATE:	

AIM:

ALGORITHM:

CODE:

Program1:

```
type: heap / copy / right / ...  
>>> print("Data Analytics Lab")
```

Output:

```
Data Analytics Lab
```

Program2:

```
>>> a=20
>>> b=10
>>> print("Addition:",a+b)
Addition: 30
>>> print("Subtraction:",a-b)
Subtraction: 10
>>> print("Multiplication:",a*b)
Multiplication: 200
>>> print("Division:",a/b)
Division: 2.0
```

Output:

```
Addition: 30
Subtraction: 10
Multiplication: 200
Division: 2.0
```

Program 3:

```
>>> name="John"
>>> dept="AIML"
>>> cgpa=8.9
>>> print("NAME:",name)
NAME: John
>>> print("DEPARTMENT:",dept)
DEPARTMENT: AIML
>>> print("CGPA:",cgpa)
CGPA: 8.9
```

Output:

```
NAME: John
DEPARTMENT: AIML
CGPA: 8.9
```

Program 4:

```
# Data Analytics Lab
## Experiment 1
### Jupyter Notebook
```

Output:

Data Analytics Lab

Experiment 1

Jupyter Notebook

Program 5:

```
import numpy as np
a = np.array([10,20,30,40,50])
print(a)
print("Mean =",np.mean(a))
print("Sum =",np.sum(a))
```

Output:

```
[10 20 30 40 50]
Mean = 30.0
Sum = 150
```

Program 6:

```
import pandas as pd
data = {
    'Name': ['John', 'David', 'Alex'], 'Marks': [85,90,95]
}
df = pd.DataFrame(data)
print(df)
```

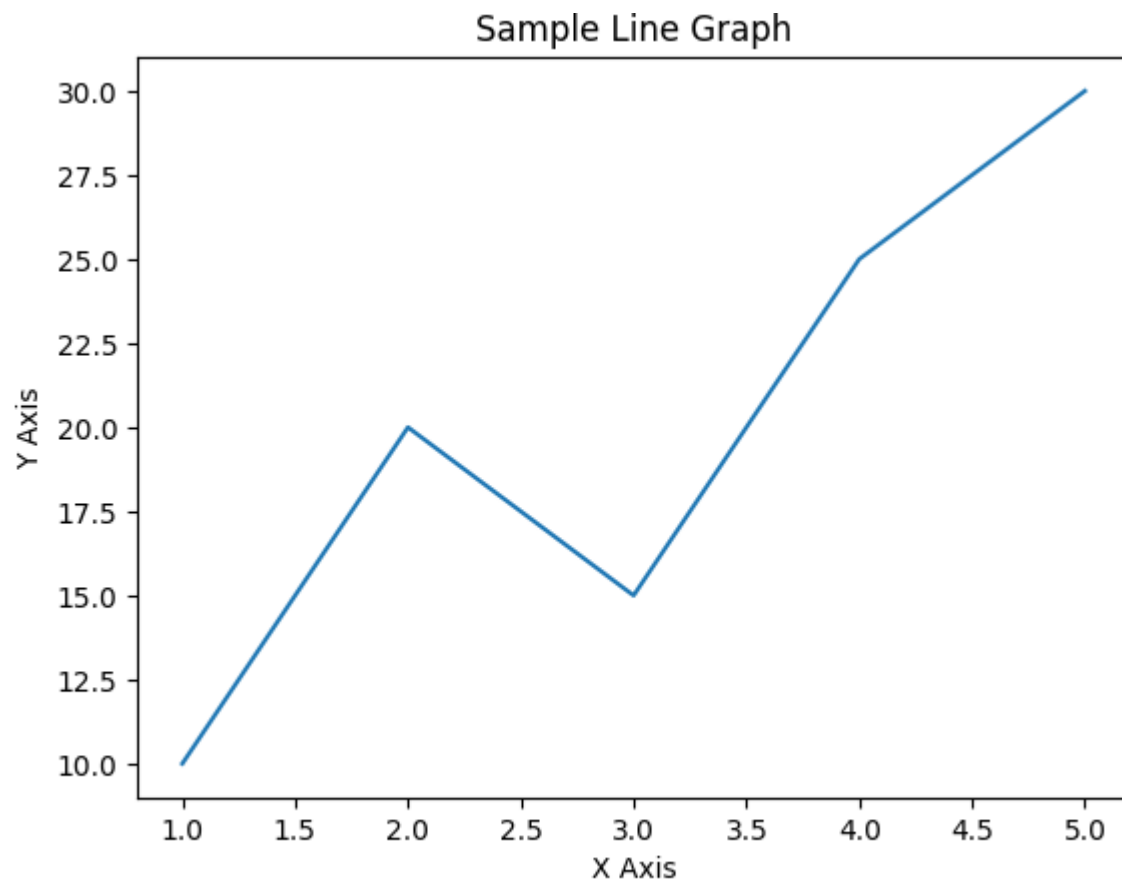
Output:

	Name	Marks
0	John	85
1	David	90
2	Alex	95

Program 7:

```
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]
plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

Output:



RESULT: